

Paramount Diagnostic Lab

Test Details

METHODOLOGY

Immunologic

RESULT TURNAROUND TIME

1 - 2 days

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Use

Quantitation of C_4 is used to detect individuals with inborn deficiency of this factor or those with immunologic disease in whom hypercatabolism of complement causes reduced levels. These diseases include lupus erythematosus, serum sickness, certain glomerulonephritides, chronic active hepatitis, and others.

Custom Additional Information

C₄ is used only by the classical pathway, so that it is decreased only when this arm is activated. In diseases activating the alternate pathway alone, C₄ levels will be normal. Total hemolytic activity (CH₅₀), C₃, and C₄ are frequently decreased in a variety of conditions producing immune complexes. C₄ levels are sensitive indicators of lupus disease activity. In hereditary angioedema, the lack of C₁ esterase inhibitor allows unopposed lysis of C₂ and C₄ by C₁ esterase, so C₄ levels will be low. C₄ deficiency has been described in association with a clinical SLE-like disease but with absence of LE cells and variable immunoglobulin or C₃ deposits in the skin biopsy, and with Henoch-Schönlein purpura or glomerulonephritis. The condition is inherited as an autosomal recessive trait with close HLA linkage. Hereditary C₄ deficiency has been associated with an increased incidence of pyogenic bacterial infections.

Specimen Requirements

Specimen

Serum (preferred) **or** plasma

Volume

1 mL (adult), 0.5 mL (pediatric)

Container

Red-top tube, gel-barrier tube, green-top (lithium heparin) tube, **or** lavender-top (K₃-EDTA) tube

Collection Instructions

Allow serum to clot at room temperature for 15 to 30 minutes. Remove serum or plasma after centrifugation and transfer to a plastic transport tube.

Stability Requirements

Temperature	Period
Room temperature	2 days
Refrigerated	8 days
Frozen	3 months
Freeze/thaw cycles	Stable x1

Reference Range

See table.

Age	Male (mg/dL)	Female (mg/dL)
0 to 30 d	Not established	Not established
1 m to 17 y	10–34	10–34
>17 y	12–38	12–38

Storage Instructions

Maintain specimen at room temperature.

Causes for Rejection

Excessive lipemia (>250 mg/dL) and rheumatoid factors (>200 IU/mL) significantly increase the apparent C_4 concentration.
